Non-Markovian exceptional points in waveguide quantum electrodynamics

Core message

Thesis

Exceptional points can arise directly from *memory*: not from a finite-dimensional instantaneous Hamiltonian or Liouvillian, but from the spectral singularities of a delay/memory evolution operator.

In Longhi's waveguide-QED realization:

- retardation produces a delay-differential equation for the excited-state amplitude;
- the relevant spectrum is the set of Laplace poles s_n;
- a non-Markovian EP occurs when two or more poles coalesce;
- experimentally visible signature: transition from monotonic relaxation to oscillatory relaxation, with real zeros of $a(t)$.

Generic memory-kernel formulation

Start from a nonlocal equation

$$\dot{x}(t) = \int_0^t K(t - t')x(t')\, dt'. \tag{1}$$

Taking the Laplace transform gives

$$\left[sI - \widetilde{K}(s)\right] \tilde{x}(s) = x(0), \tag{2}$$

where

$$\widetilde{K}(s) = \int_0^\infty e^{-st} K(t)\, dt. \tag{3}$$

The characteristic equation is

$$D(s) \equiv \det\left[sI - \widetilde{K}(s)\right] = 0. \tag{4}$$

The roots s_n are the generalized dynamical eigenvalues.

Definition of a non-Markovian EP

For simple poles,

$$\tilde{x}(s) \sim \sum_n \frac{r_n}{s - s_n} \quad \Longrightarrow \quad x(t) \sim \sum_n r_n e^{s_n t}. \tag{5}$$

At an order-p EP:

$$D(s_{\mathrm{EP}}) = D'(s_{\mathrm{EP}}) = \cdots = D^{(p-1)}(s_{\mathrm{EP}}) = 0. \tag{6}$$

Equivalently, p simple poles merge into a pth-order pole:

$$\tilde{x}(s) \sim \frac{c_{p-1}}{(s - s_{\mathrm{EP}})^p} + \cdots + \frac{c_0}{s - s_{\mathrm{EP}}}. \tag{7}$$

Therefore

$$x(t) \sim \left(c_0 + c_1 t + \cdots + c_{p-1} t^{p-1}\right) e^{s_{\mathrm{EP}} t}. \tag{8}$$

Connection to pseudomodes

If the kernel can be represented or approximated by exponentials,

$$K(t) = \sum_{j=1}^{N} \gamma_j e^{-\lambda_j t}, \tag{9}$$

introduce auxiliary variables

$$y_j(t) = \int_0^t \gamma_j e^{-\lambda_j (t-t')} x(t')\, dt'. \tag{10}$$

Then

$$\dot{x}(t) = \sum_j y_j(t), \tag{11}$$

$$\dot{y}_j(t) = -\lambda_j y_j(t) + \gamma_j x(t). \tag{12}$$

Interpretation

The auxiliary variables are a mathematical analogue of pseudomodes: the non-Markovian memory is embedded into an enlarged Markovian linear system.

Relation to Lin et al. (Nature Communications 2025)

Lin et al. formulate non-Markovian quantum EPs using extended Liouvillians generated by

- pseudomode equations of motion (PMEOM), and
- hierarchical equations of motion (HEOM).

The common structure is:

$$\text{memory kernel} \quad \Longleftrightarrow \quad \text{auxiliary modes / enlarged generator.} \tag{13}$$

Longhi's contribution is a waveguide-QED delay realization in which the same logic becomes analytically transparent:

$$\text{delay equation} \rightarrow \text{Lambert-}W \text{ pole spectrum} \rightarrow \text{pole coalescence EP.} \tag{14}$$

Physical platform: a two-point giant atom

Consider a two-level emitter coupled to a 1D waveguide at two spatially separated points $x_1 = 0$, $x_2 = d$.

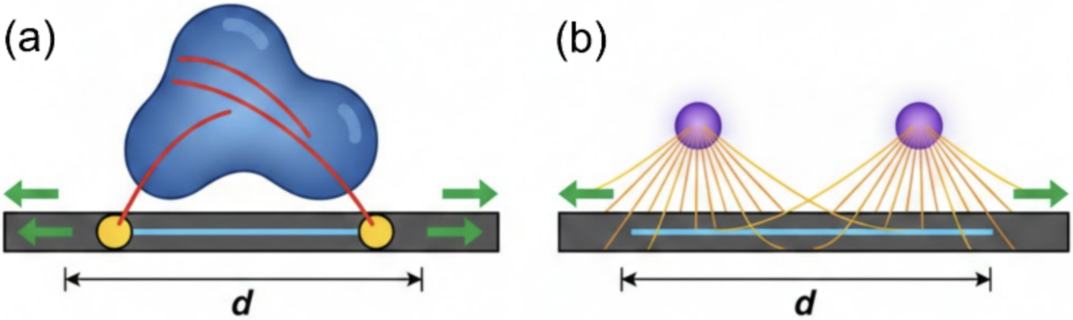
Hamiltonian in the rotating-wave approximation:

$$H = \omega_0 \sigma_+ \sigma_- + \int_{-\infty}^{\infty} dk\, \omega_k a_k^\dagger a_k + \int_{-\infty}^{\infty} dk\, \left[g(1 + e^{ikd}) \sigma_+ a_k + \mathrm{H.c.} \right]. \tag{15}$$

Single-excitation ansatz:

$$|\Psi(t)\rangle = a(t) e^{-\mathrm{i}\omega_0 t} |e, \mathrm{vac}\rangle + \int dk\, \phi_k(t) e^{-\mathrm{i}\omega_k t} |g, 1_k\rangle . \tag{16}$$

Physical platform: a two-point giant atom



Delay equation for spontaneous emission

Eliminating the waveguide amplitudes gives a memory equation. For linear dispersion $\omega_k = v_g|k|$:

$$\dot{a}(t) = -\gamma a(t) - \gamma e^{\mathrm{i}\phi} a(t-\tau)\Theta(t-\tau). \tag{17}$$

Parameters:

$$\tau = \frac{d}{v_g}, \qquad \phi = \omega_0 \tau. \tag{18}$$

Meaning of the two terms:

- $-\gamma a(t)$: instantaneous spontaneous emission from the coupling points;
- $-\gamma e^{\mathrm{i}\phi} a(t-\tau)$: coherent delayed self-feedback from photon propagation between points.

Non-Markovianity source

The atom interacts with its own past radiation.

Exact time-domain recurrence solution

The delay equation admits the recurrence solution

$$a(t) = \sum_{n=0}^{\infty} \frac{[-\gamma e^{\mathrm{i}\phi}(t - n\tau)]^n}{n!} \, e^{-\gamma(t-n\tau)}\Theta(t - n\tau). \tag{19}$$

Interpretation:

- $n = 0$: no delayed return;
- $n = 1$: one delayed feedback event;
- $n > 1$: multiple reabsorption/reemission loops.

This representation is physically intuitive, but the EP structure is cleaner in Laplace space.

Laplace spectrum and Lambert-W branches

Laplace transforming the delay equation yields

$$\tilde{a}(s) = \frac{1}{s + \gamma + \gamma e^{\mathrm{i}\phi} e^{-s\tau}}. \tag{20}$$

Thus the poles solve

$$D(s) = s + \gamma + \gamma e^{\mathrm{i}\phi} e^{-s\tau} = 0. \tag{21}$$

Equivalently,

$$(s+\gamma)\tau e^{(s+\gamma)\tau} = -\gamma\tau e^{\gamma\tau + \mathrm{i}\phi}. \tag{22}$$

Therefore

$$s_n = -\gamma + \frac{1}{\tau} W_n\left(-\gamma\tau e^{\gamma\tau+\mathrm{i}\phi}\right), \qquad n \in \mathbb{Z}. \tag{23}$$

Two-point giant atom: EP condition

An EP requires

$$D(s) = 0, \qquad D'(s) = 0. \tag{24}$$

For

$$D(s) = s + \gamma + \gamma e^{\mathrm{i}\phi} e^{-s\tau}, \tag{25}$$

we have

$$D'(s) = 1 - \gamma\tau e^{\mathrm{i}\phi} e^{-s\tau}. \tag{26}$$

Combining the two conditions gives

$$\gamma\tau e^{1+\gamma\tau+\mathrm{i}\phi} = 1. \tag{27}$$

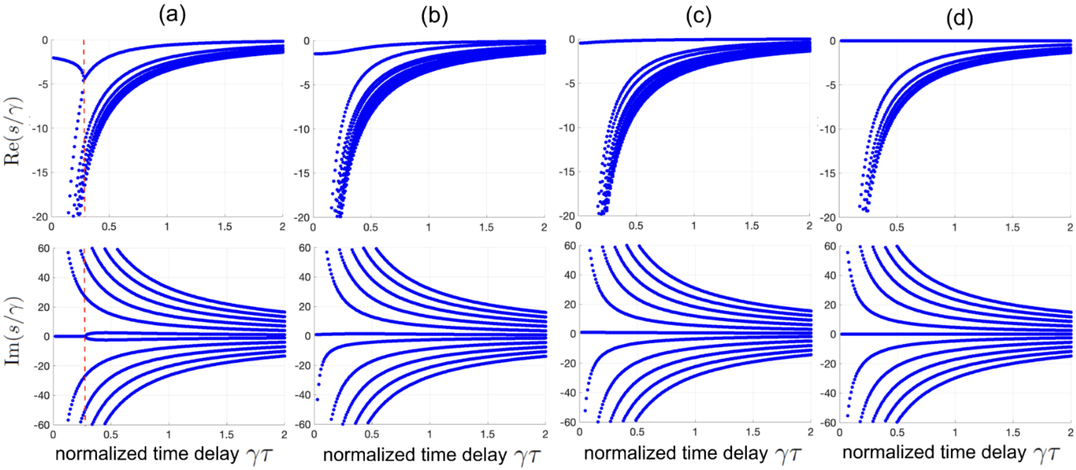
Hence the second-order EP occurs in the superradiant phase

$$\phi = 0 \pmod{2\pi}, \tag{28}$$

with critical delay

$$\boxed{\tau_{\mathrm{EP}} \simeq \frac{0.27846}{\gamma}.} \tag{29}$$

Two-point giant atom: EP condition



Dynamical transition across the EP

Below the EP:

$$\tau < \tau_{\mathrm{EP}} : \qquad s_1, s_2 \in \mathbb{R}_{<0} \tag{30}$$

so the dominant relaxation is overdamped and monotonic.

Above the EP:

$$\tau > \tau_{\mathrm{EP}} : \qquad s_\pm = -\Gamma \pm \mathrm{i}\Omega \tag{31}$$

so the decay becomes oscillatory.

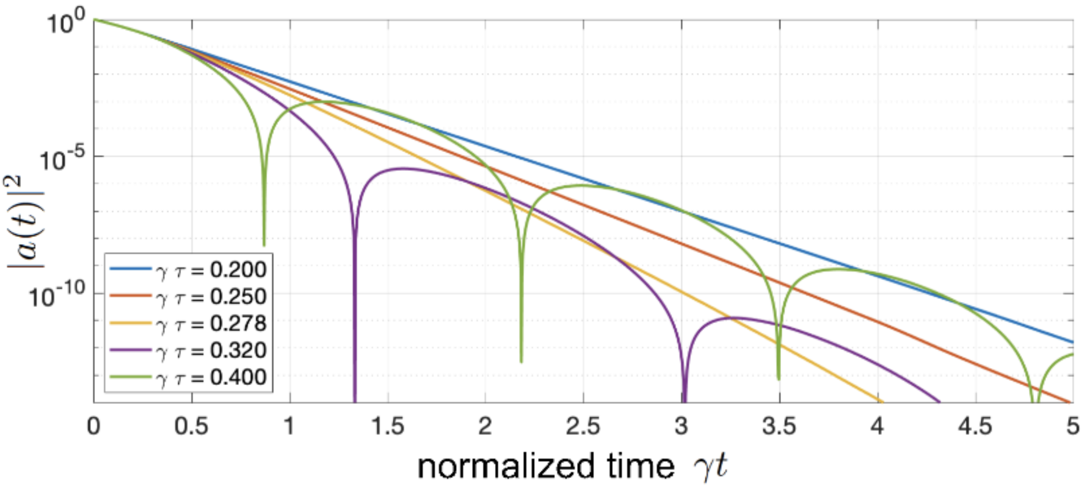
Near the EP:

$$\Omega \propto \sqrt{\tau - \tau_{\mathrm{EP}}}, \qquad T = \frac{2\pi}{\Omega} \sim (\tau - \tau_{\mathrm{EP}})^{-1/2}. \tag{32}$$

At the EP:

$$a(t) \sim t e^{s_{\mathrm{EP}} t}. \tag{33}$$

Dynamical transition across the EP



Dominant-pole pseudomode reduction

Near the EP, retain the two dominant poles s_1, s_2:

$$\tilde{a}(s) \simeq \frac{A}{(s - s_1)(s - s_2)}. \tag{34}$$

This implies

$$\ddot{a} - (s_1 + s_2)\dot{a} + s_1 s_2 a = 0. \tag{35}$$

Introduce an auxiliary variable, e.g.

$$\boldsymbol{x} = \begin{pmatrix} a \\ b \end{pmatrix}, \tag{36}$$

with first-order dynamics

$$\dot{\boldsymbol{x}} = \begin{pmatrix} 0 & 1 \\ -s_1 s_2 & s_1 + s_2 \end{pmatrix} \boldsymbol{x}. \tag{37}$$

At $s_1 = s_2$, this 2×2 matrix has a Jordan block.

What the pseudomode picture buys you

The delay system has infinitely many Lambert-W poles. The pseudomode reduction says:

$$\text{dominant poles} \quad \leftrightarrow \quad \text{effective auxiliary modes.} \tag{38}$$

Thus the non-Markovian EP can be viewed in two equivalent ways:

1. as coalescence of poles of the delay resolvent;
2. as an ordinary defective matrix in an enlarged, approximate Markovian embedding.

This is the conceptual bridge to PMEOM/HEOM non-Markovian QEP frameworks.

EP oscillations versus bound-state oscillations

The paper separates two mechanisms:

Mechanism	Pole structure	Dynamics
Non-Markovian EP	coalescing damped poles	damped oscillatory relaxation
Atom-photon bound state	pole on imaginary axis	persistent oscillations / trapping

For the two-point giant atom:

- EP appears in the superradiant phase $\phi = 0$;
- bound state appears in the subradiant phase $\phi = \pi$;
- the two mechanisms do not coexist in this minimal model.

N coupling points: multi-delay equation

For a giant atom coupled at N points $x_j = (j-1)d$ with couplings g_j, the amplitude obeys

$$\dot{a}(t) = -\sum_{m=0}^{N-1} \kappa_m e^{\mathrm{i}m\phi} a(t - m\tau)\Theta(t - m\tau), \tag{39}$$

where

$$\kappa_m = \sum_{|j-\ell|=m} \frac{\pi g_j g_\ell}{v_g}. \tag{40}$$

The Laplace-domain characteristic equation is

$$D(s) = s + \sum_{m=0}^{N-1} \kappa_m e^{\mathrm{i}m\phi} e^{-sm\tau} = 0. \tag{41}$$

An order-p EP requires

$$D(s_{\mathrm{EP}}) = D'(s_{\mathrm{EP}}) = \cdots = D^{(p-1)}(s_{\mathrm{EP}}) = 0. \tag{42}$$

Higher-order EP engineering

With N coupling points there are N delay weights $\kappa_0, \dots, \kappa_{N-1}$.

For fixed s_{EP}, the order-N conditions

$$D^{(q)}(s_{\mathrm{EP}}) = 0, \qquad q = 0, 1, \dots, N-1, \tag{43}$$

form a Vandermonde-like linear system for the effective delay weights κ_m.

The physical couplings g_j are then reconstructed from

$$\kappa_0 = \frac{\pi}{v_g} \sum_{j=1}^{N} g_j^2, \qquad \kappa_m = \frac{2\pi}{v_g} \sum_{j=1}^{N-m} g_j g_{j+m}. \tag{44}$$

Main engineering statement

For sufficiently long delay, one can design an order-N non-Markovian EP by choosing the contact-point couplings.

Example: third-order EP for $N = 3$

For three coupling points,

$$D(s) = s + \kappa_0 + \kappa_1 e^{\mathrm{i}\phi} e^{-s\tau} + \kappa_2 e^{2\mathrm{i}\phi} e^{-2s\tau}. \tag{45}$$

A third-order EP requires

$$D(s_{\mathrm{EP}}) = D'(s_{\mathrm{EP}}) = D''(s_{\mathrm{EP}}) = 0. \tag{46}$$

For $\phi = 0$, one explicit example is

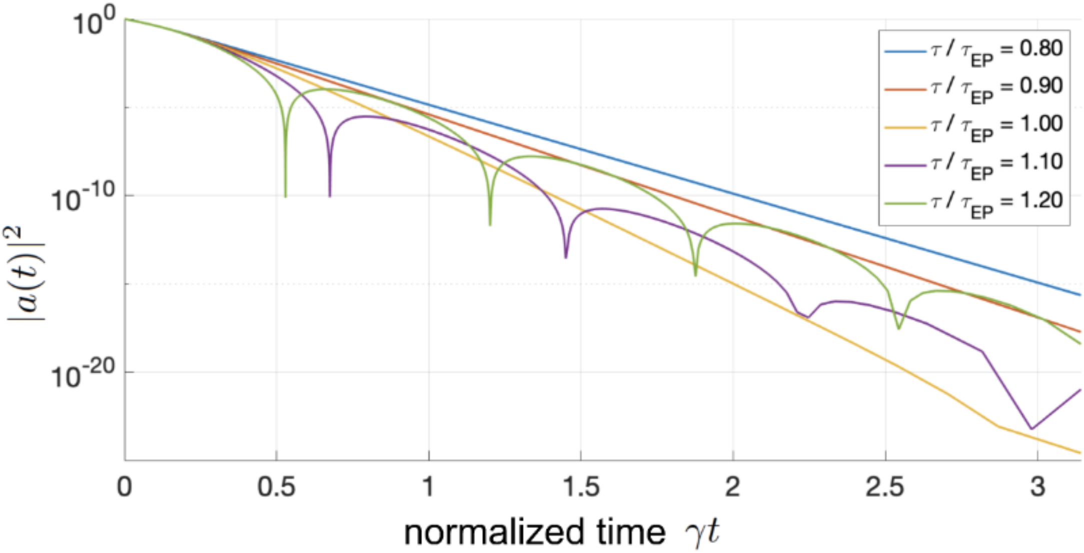
$$g_2 = g_1, \qquad g_3 \simeq -0.03846\, g_1, \tag{47}$$

with

$$\tau_{\mathrm{EP}} \simeq \frac{0.1665}{\gamma}, \qquad \gamma = \frac{\pi g_1^2}{v_g}. \tag{48}$$

Crossing this point produces a transition to oscillatory relaxation with fastest decay near the EP.

Example: third-order EP for $N = 3$



Take-home points

1. Non-Markovian EPs arise from coalescence of Laplace-domain poles, not necessarily from a finite-dimensional instantaneous Liouvillian.
2. Time delay in waveguide QED gives an analytically tractable route to such EPs.
3. The two-point giant atom has a second-order EP at $\gamma\tau \simeq 0.27846$ in the superradiant phase.
4. Crossing the EP produces a transition from monotonic to oscillatory relaxation.
5. Multiple coupling points allow engineered higher-order EPs.
6. Pseudomodes provide the bridge from memory kernels to ordinary non-Hermitian matrix EPs in enlarged space.

Open questions / possible discussion points

- Can one formulate a full density-matrix Liouvillian version rather than an amplitude-level single-excitation theory?
- What is the operational relation between the EP transition and standard non-Markovianity measures?
- How far can the pseudomode truncation be pushed beyond the dominant-pole regime?